Week 4 Project Logbook: Training Experiments & Evaluation

I. Hyperparameter Tuning & Dataset Experiments

(To be completed during the initial tuning phase)

- **Hyperparameter Grid / Random Search Matrix:**
  - *Learning Rates tested:*
  - *Batch Sizes tested:*
  - *Best Configuration Identified:*
- **Dataset Ratio & Augmentation Hypotheses:**
  - *50% vs 100% Data Impact:*
  - *Augmentation ON vs OFF Impact:*
- **Ablation Study Breakdown (Removing 1 key component at a time):**
  - *Component 1 removed → Metric change:*
  - *Component 2 removed → Metric change:*

II. Test Results Collection & Deep Analysis

(To be completed after running the best checkpoint on the held-out test set)

- **Evaluation Metrics Computed (vs. Paper's Baseline Table):**
  - *Metric 1 (e.g., SNR, F1):* Your Result _____ | Paper Result _____
  - *Metric 2 (e.g., PESQ, WER):* Your Result _____ | Paper Result _____
- **Performance Gap Hypothesis (If your results differ from the paper, why?):**
- **Systematic Failure Case Analysis (Group Review):**
  - *Pattern found in failed test samples:*
  - *Are there data quality issues?*

III. Practical Implementation & Verification Checklist

(Hands-on execution log for Week 4 based on your workflow plan)

- [ ] **Hyperparameter Tuning:** Conducted grid/random search varying learning rate, batch size, and loss weights; logged every run to identify the best configuration.
- [ ] **Dataset & Augmentation Experiments:** Trained models using 50% vs 100% data ratios and tested with augmentation ON vs OFF to measure model data-hunger.
- [ ] **Best Config Training & Ablation:** Re-trained the model with the top hyperparameter setup for full epochs (50-100); ran 3 ablation variants and built a results table matching the paper's ablation section.
- [ ] **Test Results Collection:** Ran the top checkpoint on the held-out test set; computed all relevant metrics and prepared a results summary CSV.
- [ ] **Model Internal Visualizations:** Generated metric bar charts, confusion matrices, or before/after spectrograms; visualized internals like feature maps, attention maps, or latent space t-SNEs for 5 easy and 5 hard samples.

IV. Implementation Notes & Code Snippets

- **Experiment / Evaluation Insights:**
- **Core Evaluation Script Snippet:**